

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Coding challenge questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. A hospital stores the body temperature readings of 12 patients in consecutive memory locations. A patient is considered to have a fever if the temperature value is greater than the fever threshold stored in memory. Develop an 8085 ALP to:
 - Count the number of patients with fever.
 - Find the highest recorded temperature.
 - Store both results in memory.

The screenshot displays the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window shows assembly code being loaded at address 2500H. The code includes instructions for loading data, moving data between registers and memory, comparing, jumping, and looping. The memory dump on the right shows the current state of memory, with addresses 2500H to 2609H. The memory dump shows that the data at 2500H is 0, and the data at 2600H is 08, and the data at 2601H is 95. The simulator status bar at the bottom indicates 'Simulator: Idle'.

Registers:

Register	Value
A	00
BC	00 00
DE	00 00
HL	25 0D
PSW	00 00
PC	42 28
SP	FF FF
Int-Reg	00

Flag:

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Assembly Code:

```

1  LXI H,2500H
2  MOV D,M
3  INX H
4
5  MOV A,M
6  MOV E,A
7
8  MVI C,0CH
9  MVI B,00H
10
11  LOOP: MOV A,M
12  CMP D
13  JC CHECKMAX
14  JZ CHECKMAX
15  INR B
16
17  CHECKMAX: MOV A,M
18  CMP E
19  JC NEXT
20  MOV E,A
21
22  NEXT: INX H
23  DCR C
24  JNZ LOOP
25
26  MOV A,B
27  STA 2600H
28
29  MOV A,E
30  STA 2601H
31
32  HLT
  
```

Memory Dump:

Address (Hex)	Address	Data
0A24	2596	0
0A25	2597	0
0A26	2598	0
0A27	2599	0
0A28	2600	8
0A29	2601	95
0A2A	2602	0
0A2B	2603	0
0A2C	2604	0
0A2D	2605	0
0A2E	2606	0
0A2F	2607	0
0A30	2608	0
0A31	2609	0

Assembler Message:

```

Line No  Assembler Message
0         Program assembled successfully
  
```

Your output should be: 2600 : 08 , 2601 : 95

Meaning of the Output

- 2600H = 08H → 8 patients have a temperature greater than the fever threshold.
- 2601H = 95H → The **highest temperature** recorded is

2. An automated irrigation system stores water levels measured every hour. If the water level falls below the minimum safe level, an alert should be generated. Write an 8085 ALP to:

- Count the number of low-water-level readings.

- Determine the lowest water level recorded.
- Store an alert flag if more than three readings are below the safe level.

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly window contains the following code:

```

1 LXI H, 2500H
2 MOV D, M
3 INX H
4
5 MOV A, M
6 MOV E, A
7
8 MVI C, 0CH
9 MVI B, 00H
10
11 LOOP: MOV A, M
12 CMP D
13 JC CHECKMAX
14 JZ CHECKMAX
15 INR B
16
17 CHECKMAX: MOV A, M
18 CMP E
19 JC NEXT
20 MOV E, A
21
22 NEXT: INX H
23 DCR C
24 JNZ LOOP
25
26 MOV A, B
27 STA 2600H
28
29 MOV A, E
30 STA 2601H
31
32 HLT

```

The registers window shows the state of the 8085 registers. The memory window shows the contents of memory locations from 2500 to 2609. The assembler message window shows "Program assembled successfully".

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly window contains the following code:

```

1 LXI H, 2700H
2 MOV D, M
3 INX H
4
5 MOV A, M
6 MOV E, A
7
8 MVI C, 0CH
9 MVI B, 00H
10
11 LOOP: MOV A, M
12 CMP D
13 JNC CHECKMIN
14 INR B
15
16 CHECKMIN: MOV A, M
17 CMP E
18 JNC NEXT
19 MOV E, A
20
21 NEXT: INX H
22 DCR C
23 JNZ LOOP
24
25 MOV A, B
26 STA 2800H
27
28 MOV A, E
29 STA 2801H
30
31 MOV A, B
32 CPI 04H
33 JC NOALERT
34
35 MVI A, 01H
36 STA 2802H
37 JMP END
38
39 NOALERT: MVI A, 00H
40 STA 2802H
41
42 END: HLT

```

The registers window shows the state of the 8085 registers. The memory window shows the contents of memory locations from 2700 to 2713. The assembler message window shows "Program assembled successfully".

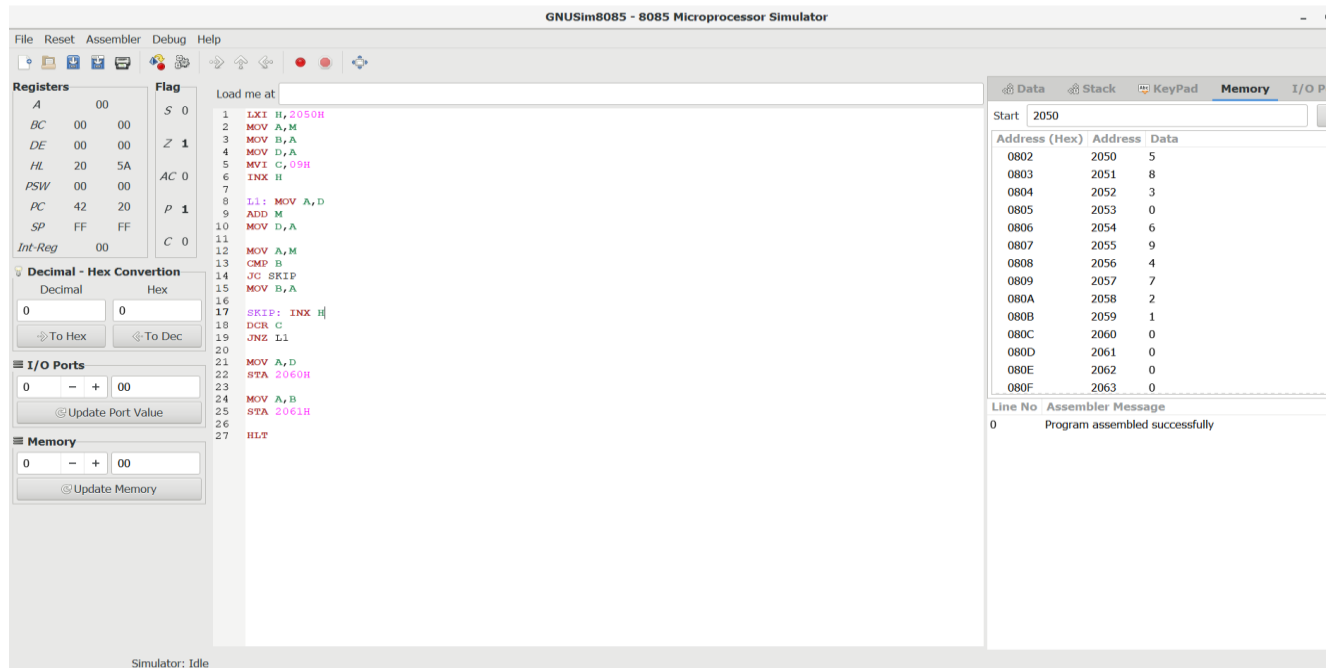
Output:

2800 : 05
2801 : 20
2802 : 01

Address	Expected Value	Meaning
2800	05	5 readings are below the safe level (40)
2801	20	Lowest water level
2802	01	Alert generated (more than 3 low readings)

3. A library stores the number of books borrowed from ten different sections. The librarian wants to identify the most frequently used section and determine the total number of books borrowed. Write an 8085 ALP to:

- Calculate the total books borrowed.
- Find the section with the highest borrowing count.
- Store both results.



Output:

Address	Data
2060	3D
2061	0A

Code of Conduct

I am Vishwa.U(192511192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

A handwritten signature in blue ink, appearing to read 'Vishwa.U', is written above the signature line.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator Faculty Incharge